07/8/26 Exp: 5 Packet Capture and Analysis using wireshark

Aim:

To capture and analyse network packets using the wireshark packet capture tool.

Theory:

Wireshark is a network protocol analyzer used to capture, filter and inspect network packets. It displays packet details, protocol information, and packet bytes. It can analyze protocols such as TCP, UDP, ARP, DNS, HTTP, ICMP and DHCP.

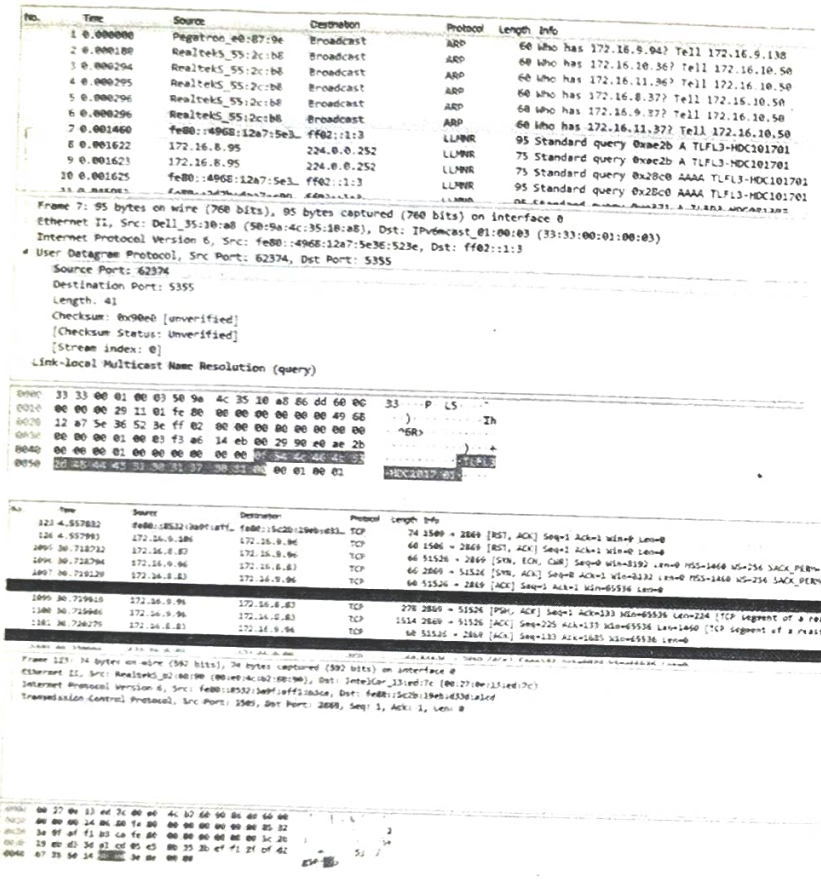
Filters used:

- TCP/UDP → tcp/udp
- ARP → arp
- DNS → dns
- HTTP → http
- ICMP → icmp
- DHCP → dhcp

Procedure:

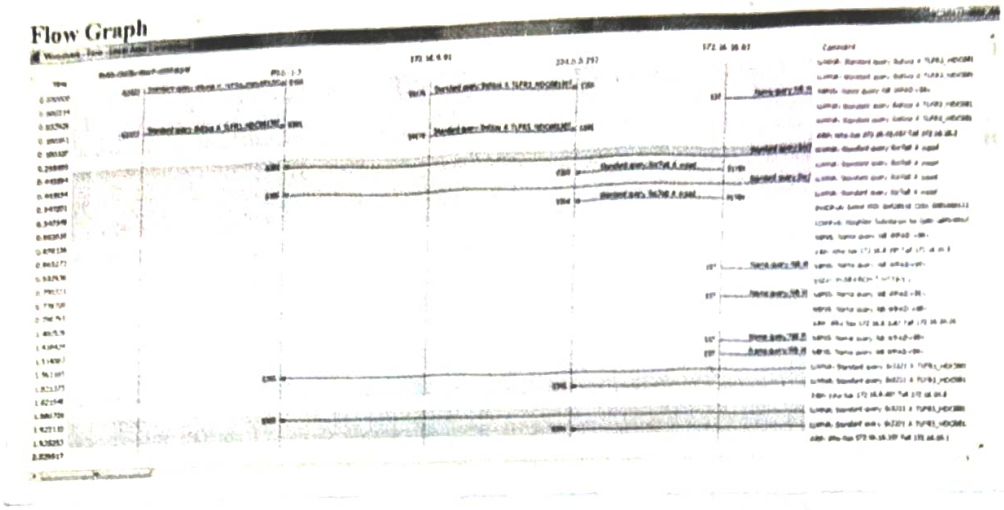
- Create a filter to display only TCP/UDP packets, inspect the packets and provide the flow graph.
- Create a filter to display only ARP and packet and inspect the packet

- Create a filter to display only DNS packets and provide the flow graph.
- Create a filter to display only HTTP packets and inspect the packets
- Create a filter to display only IP/ICMP packets and inspect the packets.
- Create a filter to display only DHCP packets and inspect the packets.





Flow Graph

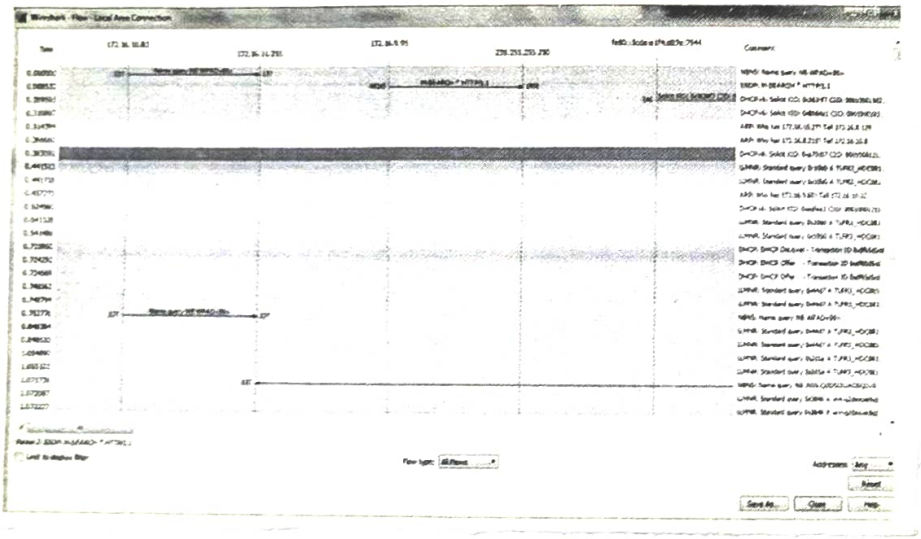


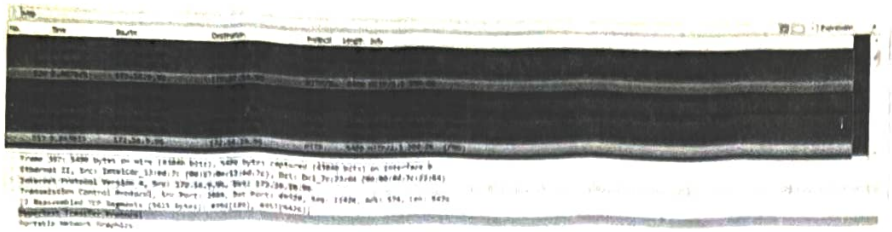
arp

No.	Time	Source	Destination	Protocol	Length	Info
6	0.255305	Foxconn_c9:c5:f0	Broadcast	ARP	60	Who has 172.16.10.15? Tell 172.16.10.3
14	0.692936	Foxconn_d0:ac:46	Broadcast	ARP	60	Who has 172.16.8.39? Tell 172.16.10.8
19	1.418424	Foxconn_c9:c9:91	Broadcast	ARP	60	Who has 172.16.8.106? Tell 172.16.10.26
24	1.880729	Foxconn_d0:ac:46	Broadcast	ARP	60	Who has 172.16.8.40? Tell 172.16.10.8
27	2.029517	Giga-Byt_92:d2:ef	Broadcast	ARP	60	Who has 172.16.10.33? Tell 172.16.10.1
41	2.509905	Giga-Byt_7c:c5:34	Broadcast	ARP	60	Who has 172.16.9.82? Tell 172.16.9.111
44	2.602358	Foxconn_c9:c8:24	Broadcast	ARP	60	Who has 172.16.8.139? Tell 172.16.10.22
46	2.743821	Dell_35:11:11	Broadcast	ARP	60	Who has 172.16.8.118? Tell 172.16.10.195
56	3.201822	Giga-Byt_92:d2:ef	Broadcast	ARP	60	Who has 172.16.10.34? Tell 172.16.10.1
60	3.237061	Giga-Byt_7c:c5:34	Broadcast	ARP	60	Who has 172.16.9.82? Tell 172.16.9.111

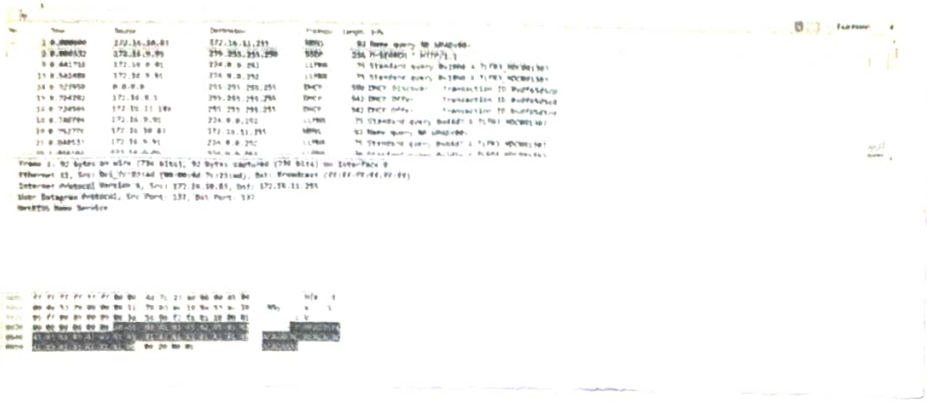
```
Frame 119: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface 0
Ethernet II, Src: IntelCor_13:ed:7c (00:27:0e:13:ed:7c), Dst: Realtek5_b2:60:90 (00:e0:4c:b2:60:90)
Address Resolution Protocol (reply)
```

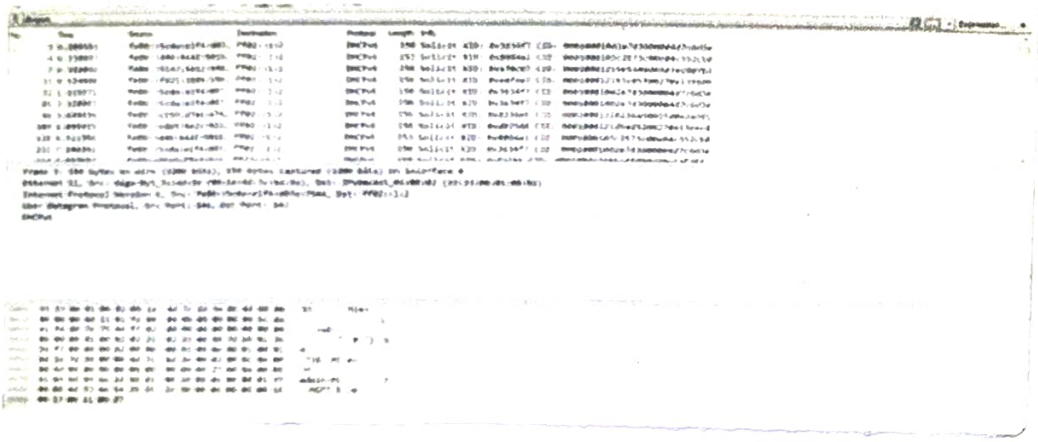
```
0000  00 e0 4c b2 60 90 00 27  0e 13 ed 7c 08 06 00 01   ··L·`··'  ···|····
0010  08 00 06 04 00 02 00 27  0e 13 ed 7c ac 10 09 c2   ·······'  ···|····
0020  00 e0 4c b2 60 90 ac 10  09 6a                      ··L·`···  ·j
```











Result :

Hence successfully captured and analysed network packets using the wireshark packet capture tool.

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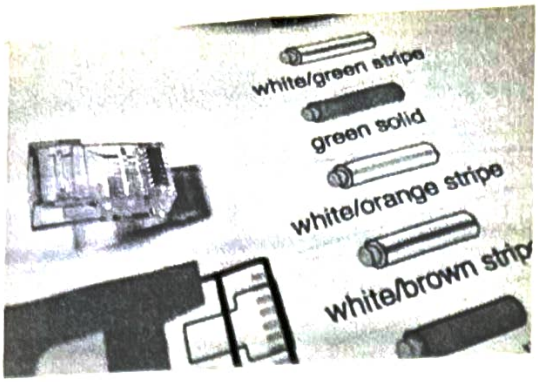
Exp: 4 Local area network using a switch and ethernet cables in your lab.

Aim:

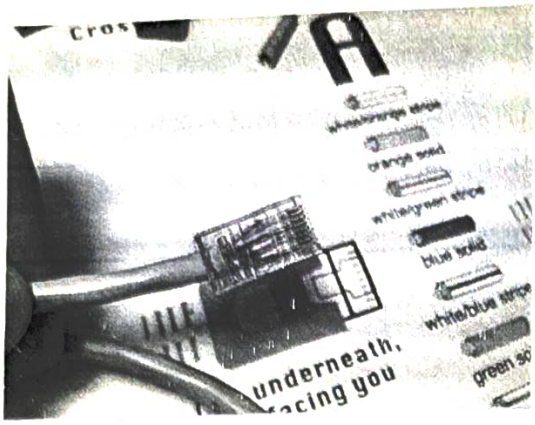
Setup and configure a LAN (Local area network using a Switch and ethernet cables in your lab.

How to setup a LAN:

- Step 1: Plan and design an appropriate network topology taking into account network requirements and equipment location.
- Step 2: You can take 4 Computer, a switch with 8, 16 and 24 ports which is sufficient for network of this size, and 4 ethernet cables.
- Step 3: Connect your computer to network switch via an ethernet cable, which is as simple as plugging one end of the ethernet cable into your computer and the other end into your network switch.
- Step 4: Check connectivity using the ping command.



- Arrange the wires evenly in order, trim any uneven ends insert them into RJ45 plug with the clip facing away and gold pins toward you



- the wires in the plug are inserted into the crimping tool and pushed down.

Result :

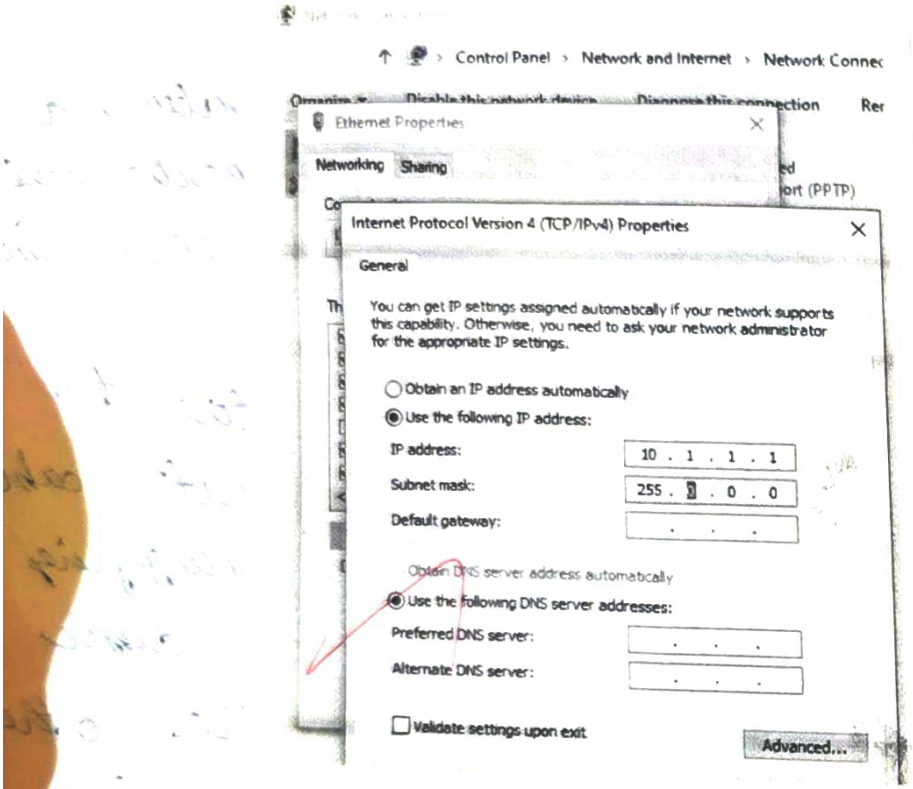
study of different types of network cables through cable and crossover is done successfully.

5) Share a folder on one computer with other computers in the LAN.

6) Access the shared folder from the other computers.

4) Share connection using with other computers in the LAN

8) Access the shared folder with other computers.



Result:

A LAN (Local area network) using a switch and ethernet cable has been setup and configured.